

Lecture 3. Summarising Data

For most of this year, will be doing two things with data: *summarising it* and *making inferences* from it. We need to report both types of statistics when conducting research:

- **Descriptive statistics**: describe and summarise your data, e.g.: What is the average score in each group of participants? How much difference (variation) is there within each group? How many participants reached a particular threshold?, etc.
- **Inferential statistics**: these require running **statistical tests** on your data to determine whether there is a significant difference between groups, whether two variables are associated, etc.

Why summarise data?

Human behaviour is complex and determined by a multitude of factors, so we cannot rely on individual case studies or observations to make inferences about its causes. That is why in psychology we are often interested in general trends or averages across multiple observations. This requires the collection of large amounts of data from many participants, which can be hard to make sense of at a glance – hence, we need a way to distil complex sets of data and describe them for the purposes of:

- Generalisation – what are undergraduate psychology students like in general?
- Prediction – what will the typical student in this class next year be like?

Frequency distributions

- **Absolute frequency** = how many?
- **Relative frequency** = what proportion?

Question: “Have you ever studied computer coding?”

(i)	Absolute Frequency (j)	Relative Frequency	Percentage
Yes	74	$74/353 = 0.21$	21 %
No	279	$279/353 = 0.79$	79 %

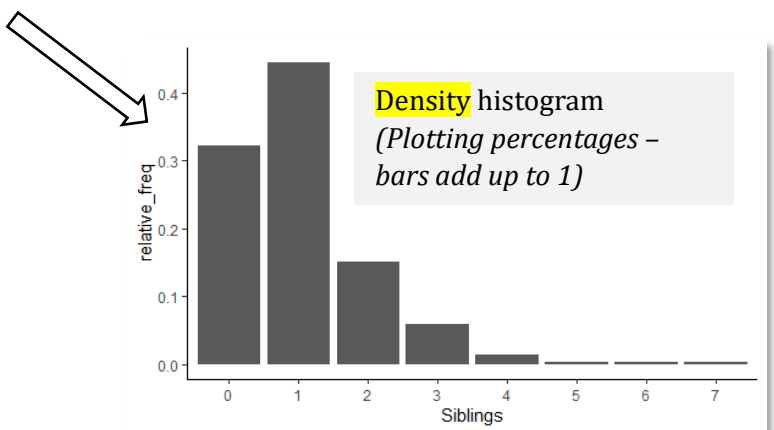
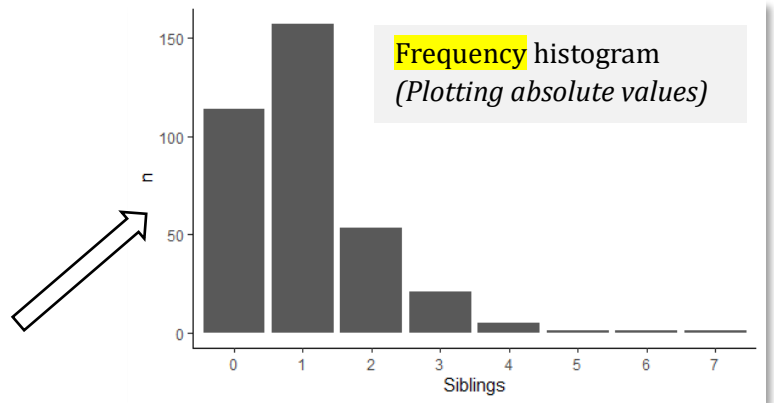
$$\text{Relative frequency}_i = \frac{\text{Absolute Frequency}_i}{\sum_{j=1}^N \text{Absolute Frequency}_j}$$

Visualising frequencies

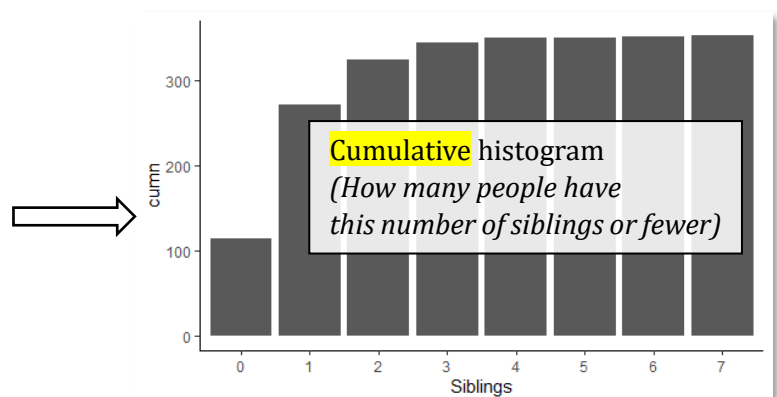
Histogram = a graph used for visualising frequencies. Helpful when you have more than 2-3 categories in the variable of interest:

Question: "How many siblings do you have?"

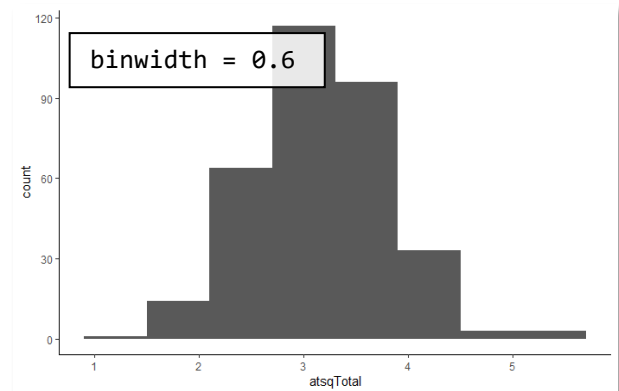
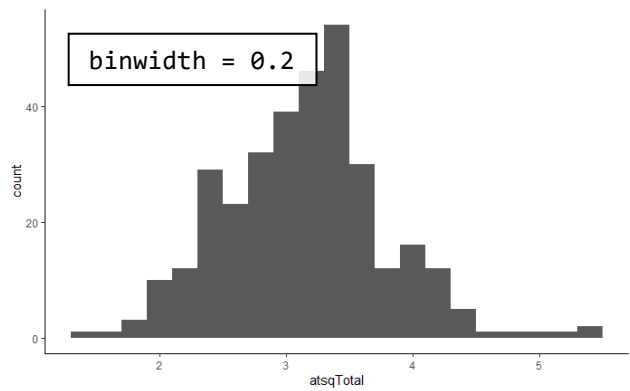
Number of siblings	Absolute Frequency	Relative Frequency
0	114	0.32
1	157	0.44
2	53	0.15
3	21	0.06
4	5	0.014
5	1	0.003
6	1	0.003
7	1	0.003



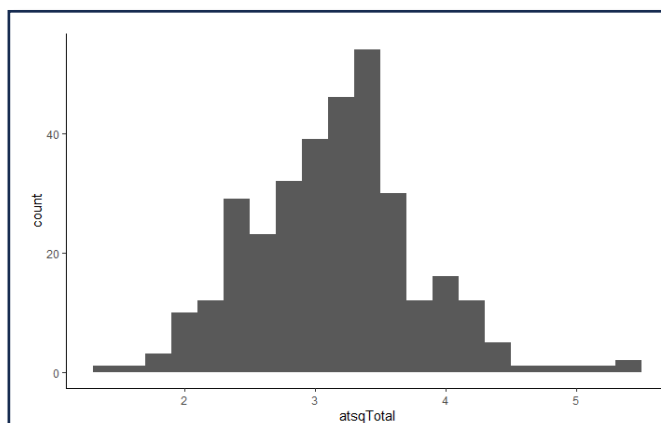
Number of siblings	Absolute Frequency	Cumulative Frequency
0	114	114
1	157	271
2	53	324
3	21	345
4	5	350
5	1	351
6	1	352
7	1	353



The above histograms all visualise discrete quantities (e.g., each bar represents a concrete number of siblings). However, some variables (interval and ratio data) can take lots of different values that don't correspond to neat independent bars. We have to group such values together into **bins**, where each bin represents a range of values:



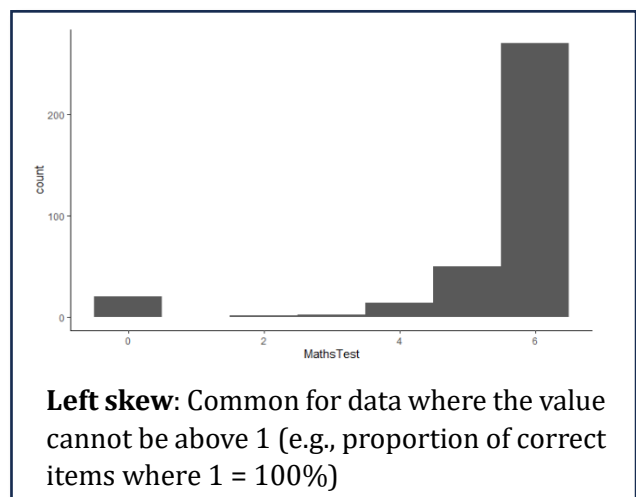
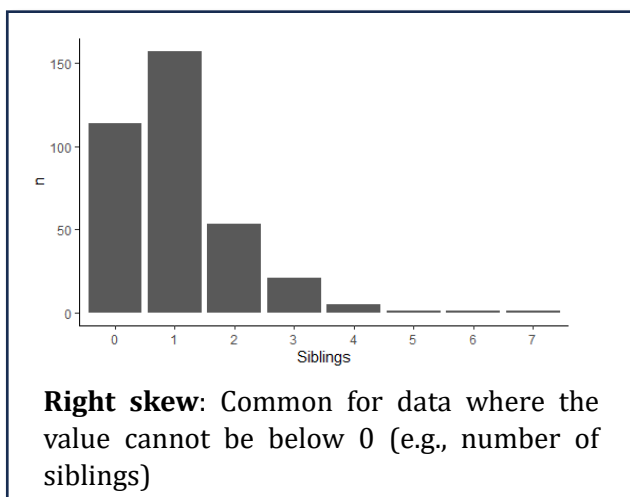
Symmetrical and skewed distributions



Symmetrical (**normal**) distribution:

- Scores are distributed around the mid-point
- Common for numeric data (e.g., questionnaire scores)

For psychological measures, skew can be indicative of **floor/ceiling effects** – when a task is too hard/ easy to capture variation in ability. Example: Lots of people are either getting almost all items or almost no items in the



Measures of Central Tendency: Mean and Median

Central tendency = a single value that represents the whole distribution (the average). It can be measured in different ways:

Mode

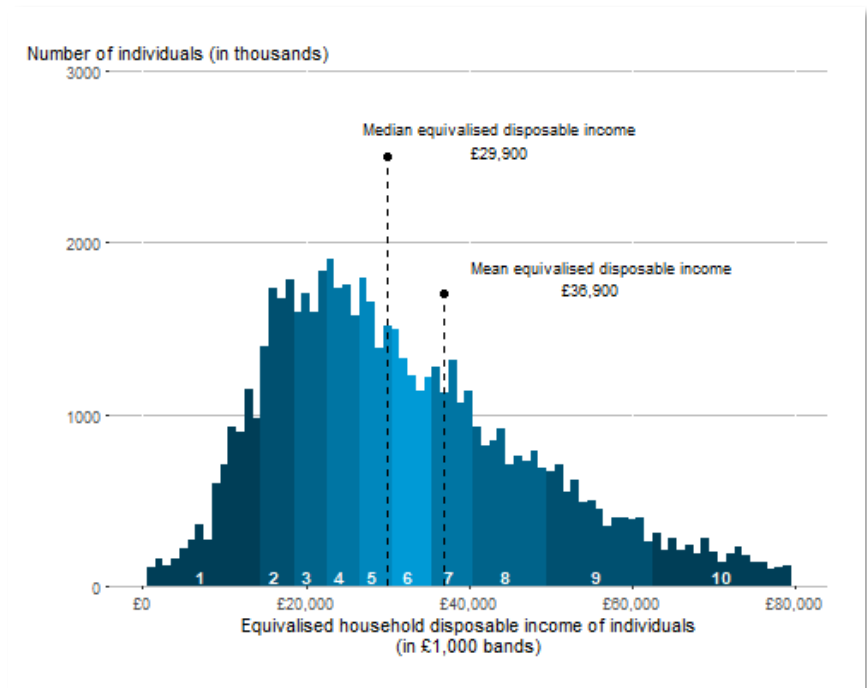
the most common value
(useful for nominal data)

Mean

the sum of values divided by
the total number of values
(downside: is sensitive to
outliers and skew – the mean
might not fall exactly where
most values are)

Median

the middle value in an
ordered set of values



Measures of Dispersion

We want some way to measure how far on average each data point is from the mean

⇒ **The Sum of Squared Differences**: Calculate each data point's distance from the mean, square them, and sum them.

Observed scores	Group Mean	Absolute differences	Squared differences
3	4.5	1.5	2.25
5	4.5	0.5	0.25
2	4.5	2.5	6.25
8	4.5	3.5	12.25

$$\sum_{i=1}^n (x_i - \mu)^2$$

$$2.25 + 0.25 + 6.25 + 12.25 = 21$$

⇒ **Variance**: Sum of squares divided by total number of observations minus one*

* (We have to subtract 1 as we have lost a degree of freedom because we are estimating the population mean from the sample mean)

$$\frac{\sum_{i=1}^n (x_i - \mu)^2}{N} \qquad \frac{2.25 + 0.25 + 6.25 + 12.25}{4 - 1} = 7$$

⇒ **Standard Deviation**: The square root of the variance. It's used to convert the variance into the original units of measurement.

$$\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{X})^2}{n - 1}} \qquad SD = \sqrt{\frac{2.25 + 0.25 + 6.25 + 12.25}{4 - 1}} = 2.65$$

Here is a simple web app demonstrating how the mean and the standard deviation of a data set change the distribution of scores. You can play with the sliders and observe how they affect the shape of the distribution:
https://msir.shinyapps.io/Mean_and_SD_app/

Descriptive statistics in R

R can generate descriptive statistics easily with the `summarise()` function.

`summarise()` makes a table of whichever descriptives you specify:

```
summarise(questionnaire_data,
  [column name] = [function],
  [column name] = [function],
  ...
)
```

The functions for descriptive statistics are `mean()` or `median()`, `sd()`, `n()`.

```
{r}
summarise(p_info, meansibs = mean(Siblings, na.rm = TRUE),
  sdsibs = sd(Siblings, na.rm = TRUE),
  minsibs = min(Siblings, na.rm = TRUE),
  maxsibs = max(Siblings, na.rm = TRUE))
```

meansibs <dbl>	sdsibs <dbl>	minsibs <dbl>	maxsibs <dbl>
1.031161	1.028924	0	7

1 row

Z-scores

Z-scores are a way of standardising individual participants' scores. They reflect where a given score lies within the distribution (how many SDs from the Mean). A z-score of 1 means the participant scored 1 Standard Deviation above the mean.

$$z = \frac{x - \bar{x}}{SD}$$

For example, if we have a sample with $M=3.15$ and $SD=.63$, then a participant who scored 4.4 can also be represented with a z-score of 1.98 – i.e., 1.98 standard deviations above the average score for the sample.

$$z = \frac{x - \bar{x}}{SD} = \frac{4.4 - 3.15}{0.63} = 1.98$$

Interpreting z-scores

If data are normally distributed:

- 68.2% of the data points can be expected to lie within one SD from the mean
- 95.4% can be expected to lie within two SDs of the mean

